

## SALES DATA ANALYSIS PROJECT (FULL GUIDE)

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### 1. Project Title

#### Sales Data Analysis and Dashboard Development

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### 2. Objective (Copy Paste)

Analyze sales data to identify trends, evaluate product performance, and understand regional sales distribution. The goal is to generate insights that support better business decisions.

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### 3. Tools Used

- Python (Pandas, Matplotlib)
  - Excel / Power BI
  - CSV Dataset
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### 4. Dataset Explanation (Copy Paste)

The dataset contains transactional sales records with the following fields:

- Order\_ID → Unique order number
  - Date → Transaction date
  - Region → Sales location
  - Category → Product category
  - Product → Product name
  - Sales → Revenue generated
  - Profit → Profit earned
  - Quantity → Units sold
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### 5. FULL PYTHON CODE (Copy Paste & Run)

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv('Dataset/powerbi_sales_data.csv')

# Show first rows
print("Sample Data:")
print(df.head())

# Total Sales & Profit
total_sales = df['Sales'].sum()
total_profit = df['Profit'].sum()

print("\nTotal Sales:", total_sales)
print("Total Profit:", total_profit)

# Sales by Region
region_sales = df.groupby('Region')['Sales'].sum()
print("\nSales by Region:\n", region_sales)

# Top Products
top_products = df.groupby('Product')['Sales'].sum().sort_values(ascending=False).head(5)
print("\nTop 5 Products:\n", top_products)

# Monthly Trend
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M')

monthly_sales = df.groupby('Month')['Sales'].sum()

# Plot 1: Region Sales
region_sales.plot(kind='bar', title='Sales by Region')
plt.xlabel("Region")
plt.ylabel("Sales")
plt.show()

# Plot 2: Monthly Trend
```

```
monthly_sales.plot(kind='line', title='Monthly Sales Trend')
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

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## 6. POWER BI DASHBOARD (What to Build)

Create these visuals:

- KPI Cards:
    - Total Sales
    - Total Profit
  - Bar Chart:
    - Region vs Sales
  - Pie Chart:
    - Product contribution
  - Line Chart:
    - Sales over time
  - Slicers:
    - Region
    - Category
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## 7. KEY INSIGHTS

- Sales performance varies significantly across different regions
  - A small group of products contributes to a large portion of revenue
  - Sales trends fluctuate over time, indicating possible seasonal patterns
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## 8. RECOMMENDATIONS (Copy Paste)

- Focus on high-performing regions to maximize revenue

- Increase promotion for top-selling products
- Analyze low-performing areas to identify improvement opportunities

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## 9. README

### # Sales Data Analysis Project

#### ## Overview

This project analyzes sales data to identify trends and generate business insights.

#### ## Tools Used

- Python (Pandas, Matplotlib)
- Power BI

#### ## Features

- Sales trend analysis
- Regional performance comparison
- Product-level insights

#### ## How to Run

1. Open Python file
2. Run analysis
3. Build dashboard in Power BI

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## 10. INTERVIEW EXPLANATION (VERY IMPORTANT)

If interviewer asks, say this:

"I worked on a sales data analysis project where I analyzed transactional data using Python and built a Power BI dashboard. I identified regional performance differences, top-selling products, and monthly trends. Based on the analysis, I suggested focusing on high-performing regions and optimizing low-performing areas."